



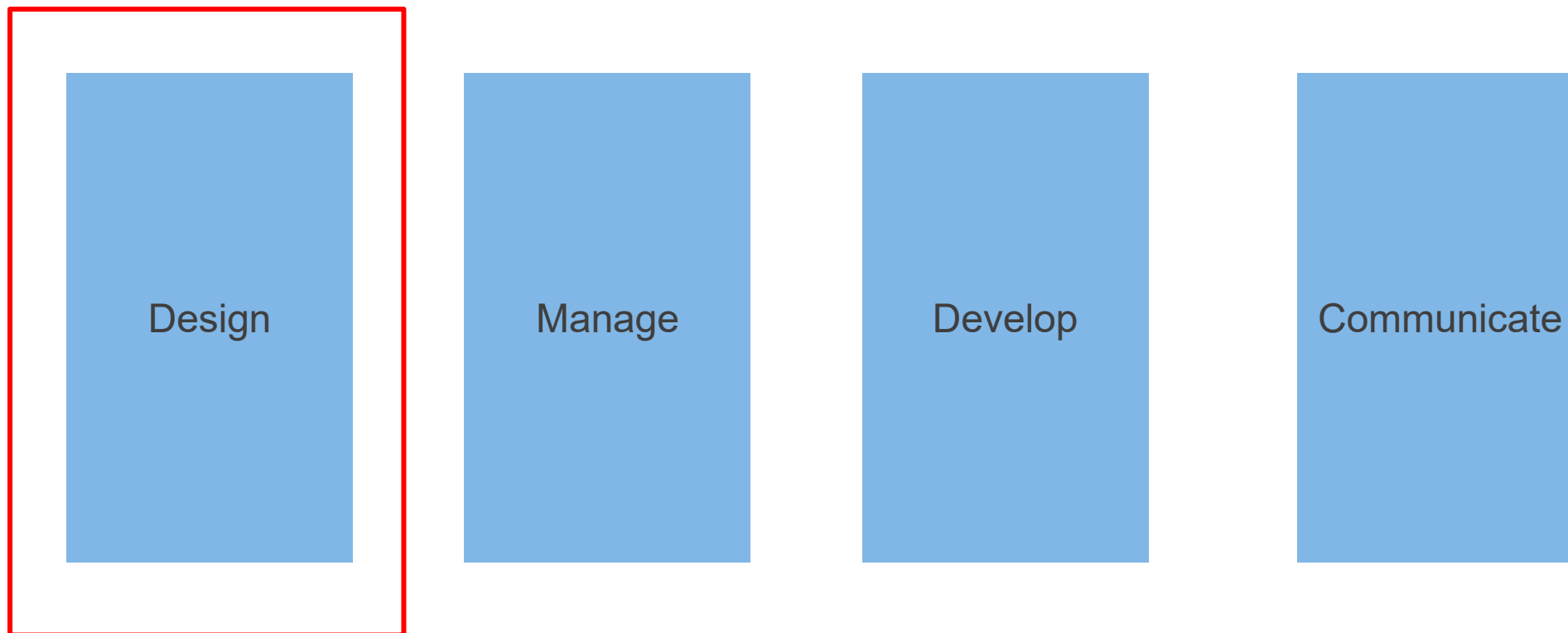
**Politecnico
di Torino**

Applied Data Science Project

L21 – Functional requirements and diagrams



Pillars





Knowledge tools

- Customer Journeys
- Persona canvas
- User requirements
- System diagrams



Knowledge tools

- Customer Journeys
- Persona canvas
- User requirements
- System diagrams

lectured by Prof. Frisiello

From user to requirements

Personas and stories are mapped into journeys, cases and requirements

1.
User goal

Goal or story level



Personas
User stories

2.
Task flow

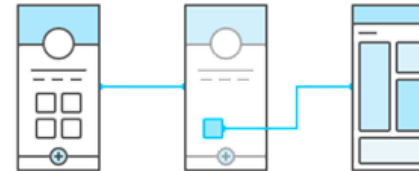
Action level



User journeys
Use cases
User requirements

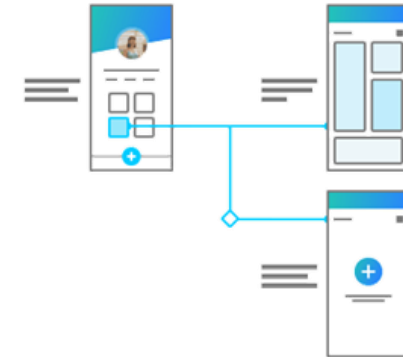
3.
Wireflow

Component level



4.
User Flow

Interaction level



UX Collective, [User flow is the new wireframe](#)

Aligning User Stories, Use Cases and Requirements

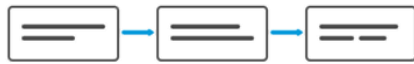
From user to requirements

From the user requirements, it is derived the system requirements

2.

Task flow

Action level



System requirements

User journeys

Use cases

User requirements

User stories

As a <user role>
I want <goal>
so that <benefit>.

[How to craft a good user story](#)

Use cases

Two alternatives

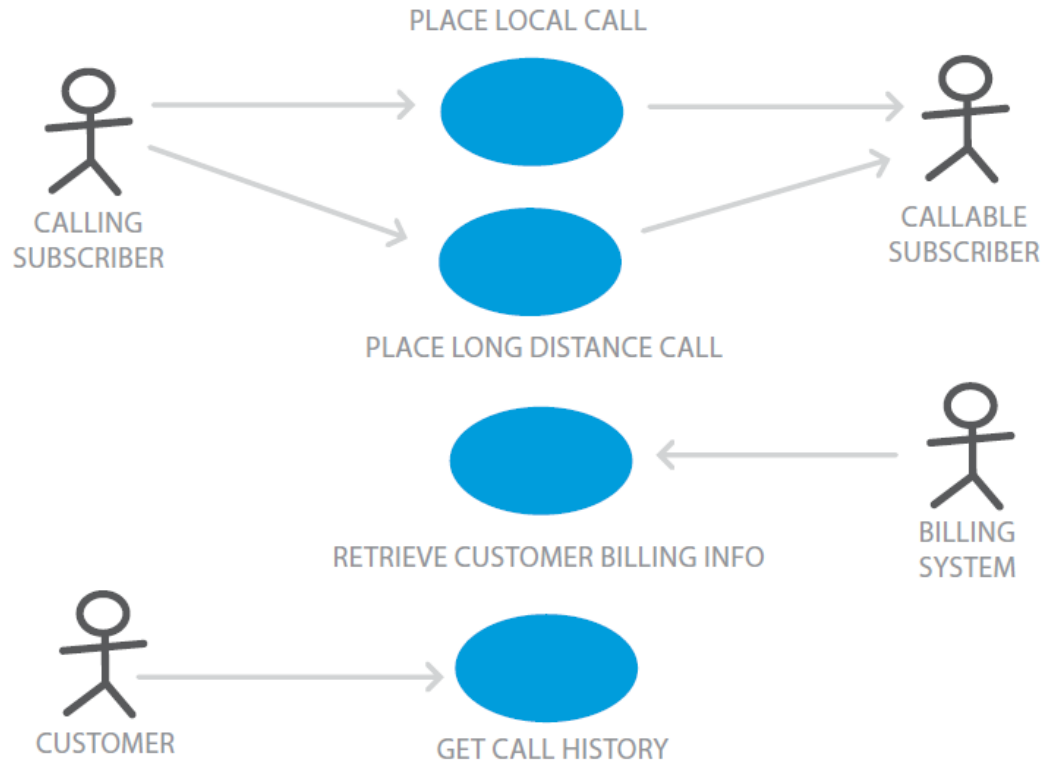


FIGURE 1: THE USE-CASE DIAGRAM FOR A SIMPLE TELEPHONE SYSTEM

BASIC FLOW

1. Insert Card
2. Validate Card
3. Select Cash Withdrawal
4. Select Account
5. Confirm Availability of Funds
6. Return Card
7. Dispense Cash

ALTERNATIVE FLOWS

- A1 Invalid Card
- A2 Non-Standard Amount
- A3 Receipt Required
- A4 Insufficient Funds in ATM
- A5 Insufficient Funds in Acct
- A6 Would Cause Overdraft
- A7 Card Stuck
- A8 Cash Left Behind
- etc..

FIGURE 2: THE STRUCTURE OF A USE-CASE NARRATIVE

Jacobsen I. (2011). [Use-case 2.0. The Guide to Succeeding with Use Cases](#)

User requirements

Functional: What does the system do?

- Capacities, characteristics, functionalities, services provided by the system
- Use verbs to name them

Non functional: How does the system work?

- Properties of the system
- Operational modes (such as robustness, precision, fairness)
- Use attributes to name them

Functional requirements

- Defined according to the results of the user requirement
- Classified according to priority
- Can be adapted in the execution of a project
- Must be validated when testing the system by the end users

ID Req	Requirement description	Status	Actions	Priority
RE01	The system offers remote onboarding	 In progress	Add voice in the authentication process	Must have



Agreed



To do



In progress



Done



Future

Recommendations to describe requirements

- Requirements are described in a positive fashion and are precise

positive

The system does not allow to access with an outdated smartphone



If the user has an outdated smartphone, the system guides the user to update



precise

The system lets users access



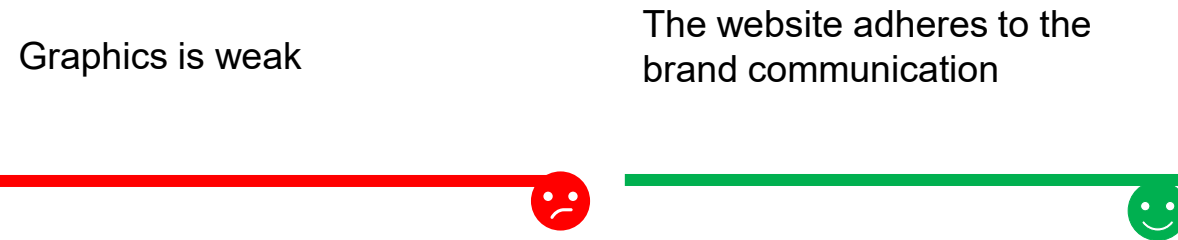
The system allows the access if a user records a video, holds an electronic ID



Garrett, Jesse James. [The elements of user experience. User-centered design for the web.](#) New Riders, 2011.

Recommendations to describe requirements

- Be objective. Requirements should be false if the conditions are not satisfied



Garrett, Jesse James. [The elements of user experience. User-centered design for the web.](#) New Riders, 2011.

Priorities

- There are things that require more attention than others
- It is recommended the Moscow Method

Mo

MUST HAVE

The most vital things you can't live without

S

SHOULD HAVE

Things you consider as important, but not vital

Co

COULD HAVE

Things that are nice to have

W

WON'T HAVE

Things that provide little to no value you can give up on

Non functional requirements

- Defined according to the results of the user requirement
- Classified according to priority
- Can be adapted in the execution of a project
- Must be validated when testing the system by the end users

ID Req	Reference personas	Requirements	Priority
RE01	Layla – Citizen	Simplified access to the municipality registry though her smartphone	Must have

Report listing requirements

1. Introduction
 - 1.1 Value Proposition
 - 1.2 Application area
 - 1.3 Background
 - 1.4 Glossary
2. Description
 - 2.1 Functionalities
 - 2.2 General constraints
 - 2.3 Use context
 - 2.4 User characteristics
3. Specific requirements
 - 3.1 Functional requirements
 - 3.2 Non functional requirements

Table 1 — Example documentation by context of use

Subclause ^a	Context of use	Could the context of use influence ease of operation?	Design limits established based on available data?	Context of use accounted for in design?
6.1	Main goal(s)			
6.2	Factors relating to other equipment			
6.3	Physical environmental factors			
6.4	Social environmental factors			
^a Refers to the breakdowns of context of use given in the subclauses of Clause 6.				

Table 2 — Example documentation by user characteristic

Subclause ^a	User characteristics	Could the user characteristic influence ease of operation?	Design limits established based on available data?	Range of user characteristics accounted for in design?
7.2.1	Cognitive abilities			
7.2.2	Knowledge and experience			
7.2.3	Cultural differences			
7.2.4	Literacy			
7.2.5	Language			
7.3.2	Body dimensions			
7.3.3	Biomechanical abilities			
7.3.4	Visual abilities			
7.3.5	Auditory abilities			
7.3.6	Handedness			
7.4.1	Demographics in general			
7.4.2	Age			
7.4.3	Gender			

Report listing requirements

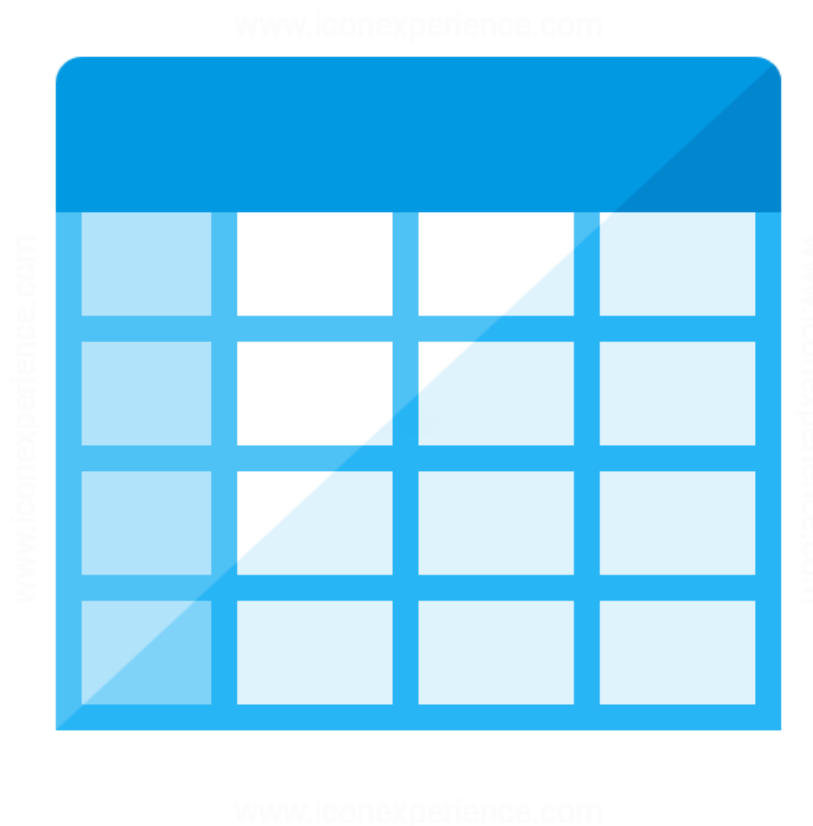
1. Introduction
 - 1.1 Value Proposition
 - 1.2 Application area
 - 1.3 Background
 - 1.4 Glossary
2. Description
 - 2.1 Functionalities
 - 2.2 General constraints
 - 2.3 Use context
 - 2.4 User characteristics
3. Specific requirements
 - 3.1 Functional requirements
 - 3.2 Nonfunctional requirements

User Story	Functional Requirements & Priorities			
	High Priority (i.e. Must have)	Medium Priority (i.e. Should have)	Low Priority (i.e. Could have)	No Priority (i.e. Won't have)
I (Angie) as the director of Latinitas would like to be able to export donor history data (name, data, amount, payment type) so that I can report to my board on increases and decreases in donor dollars.	- Create Donor Table to store info about donor - Create Donation Table used to store donations but Donor and Date. - Build report query that pulls all donors and donation info - Report must be able to run via a button on a Reports page that only admin people have access to see	- Allow report to run with filter criteria - Allow user to save preset settings on report filter criteria to speed up running reports	Have reports be automated so they run via a nightly program that emails report to admins	Reports to be accessible via mobile devices

REQ ID	ODOT PRIORITY	REQUIREMENT	AUDIENCE SEGMENTS					
			C	ICT	TT	ADA	CT	ALL
RR001	2	The system should allow the user to enter a multi-point route using a combination of the criteria specified in MP004c-f and h.	X	X	X	X		
RR0011	1	The system shall allow the user to select destination points by clicking on the map.	X	X	X	X		
		The system shall allow the user to specify the following when determining road routes (note: this functionality is for trip planning.):						
RR002a	2	- starting date AND/OR ending date if only one date is specified, the system calculates the other.			X	X		
RR002b	2	- starting time/ending time if only one time is specified, the system calculates the other.			X	X		
RR002c	2	- month of travel (instead of start/end)			X	X		
RR002d	4	- quickest route (by time)						X
RR002e	1	- shortest route (by miles)	X	X	X	X		
RR002f	2	- most scenic route (based on scenic byways within a user-specified mile radius of the direct (shortest) route)			X	X		
RR002g	3	- routes most recommended by others			X	X		

App for editing

The most used app for editing functional and nonfunctional requirements is a spreadsheet





Functional Diagram

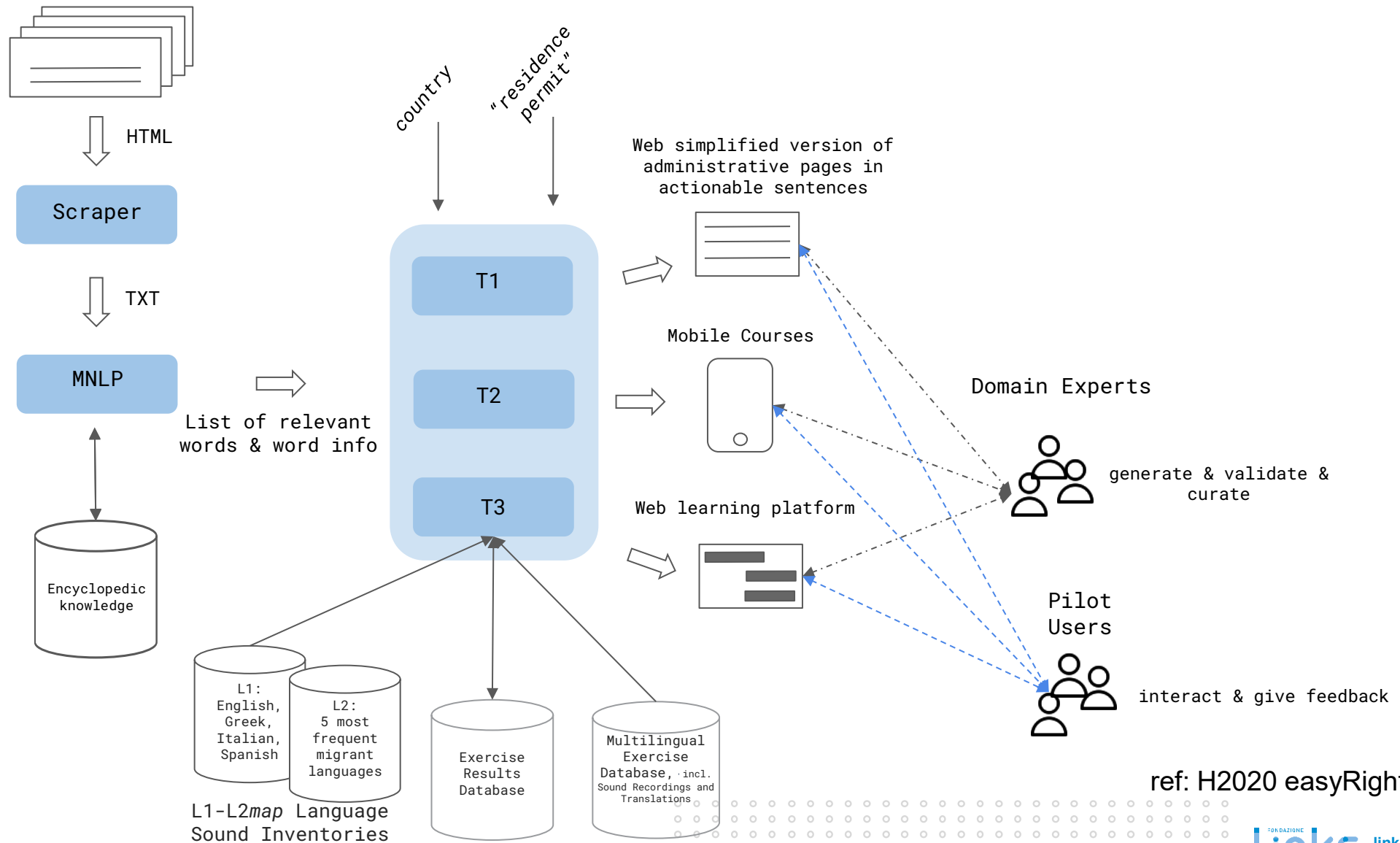
It describes the functions, links and interplay of the whole system (either be a project or a software)

Note of styles:

- functions or software components are represented by blocks
- input and output elements of a block represented with arrows lines
- relationships among functions are labelled with text
- stakeholders (such as users) are illustrated with icons



Example



ref: H2020 easyRights project

App for editing

Numerous digital tools available to create a functional diagram (many are collaborative)

In this course, we will use <https://www.diagrams.net>

- it is available both as online application running in a browser and as desktop application
- it can be utilized with a free license
- it synchronises with GDrive, OneDrive, other cloud services and locally

Diagram of Didi's tool

Hands on

Diagram of the whole project

Represent the target user (Didi's, owner of the restaurant)

Represent the data source

Add (at least) one block to define the machine intelligence that computes the sentiment

Represent the output

Diagram of the software

Represent the input text

Represent the classifier

- represent the vectorization
- represent the classification

Represent the output

Diagram of Didi's project

Whole project

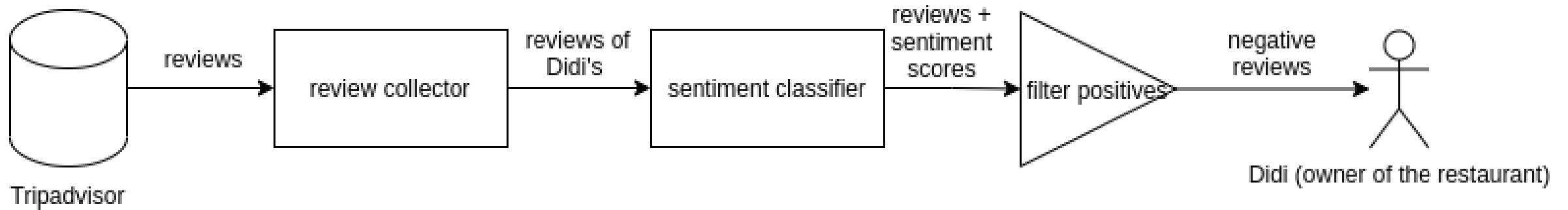
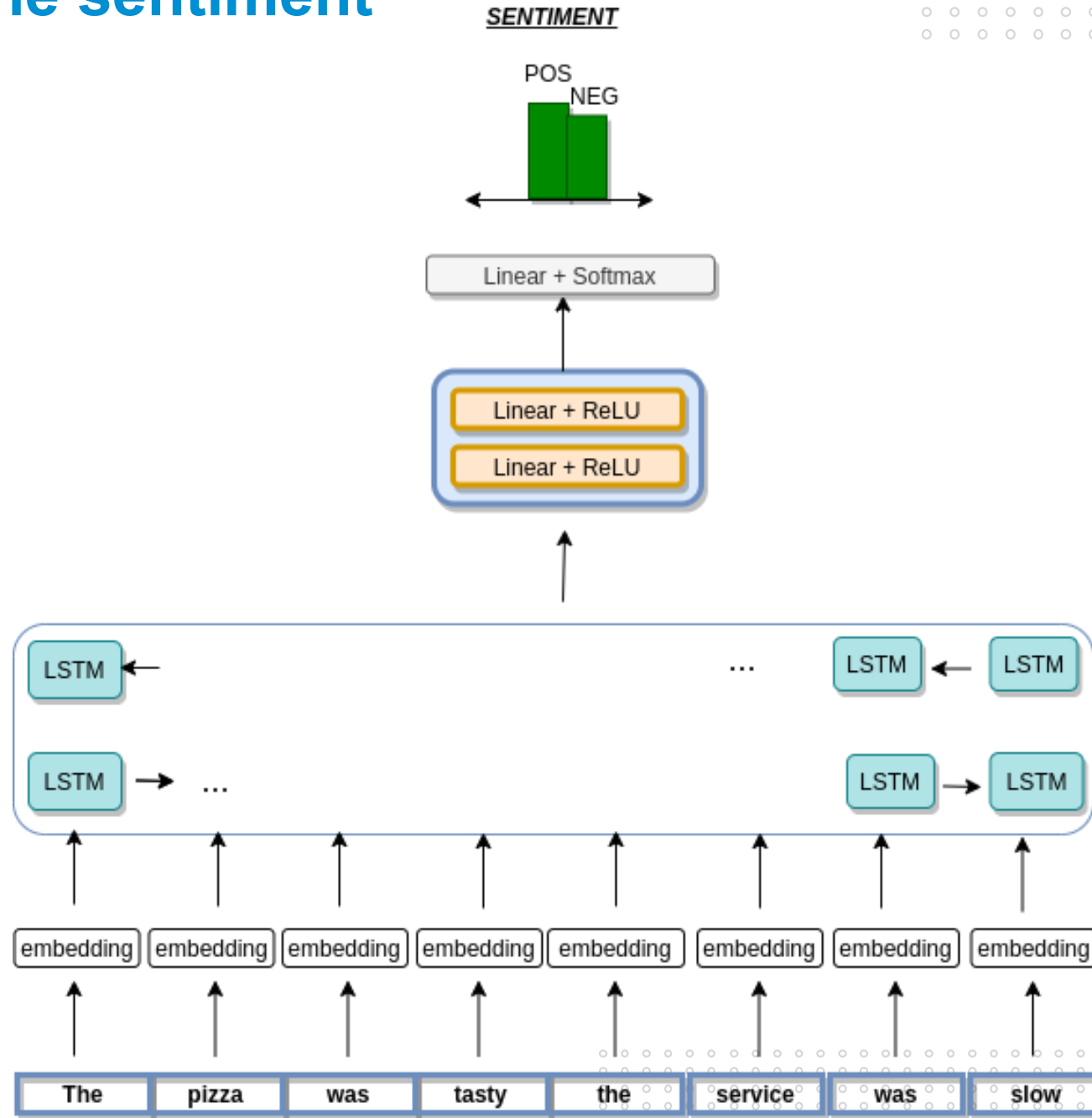


Diagram of the sentiment classifier

Software



People + AI initiative

Inspiring reading

People + AI Guidebook

<https://pair.withgoogle.com/guidebook>





Thank you for your attention.

Questions?



CONTACTS

Giuseppe Rizzo

Adjunct Professor (Politecnico di Torino)
Program Manager (LINKS Foundation)

giuseppe.rizzo@polito.it

FONDAZIONE LINKS
Via Pier Carlo Boggio 61 | 10138 Torino
P. +39 011 22 76 150
LINKSFOUNDATION.COM